



Cambridge (CIE) IGCSE Computer Science



Your notes

Digital Currency

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What is digital currency?

- Digital currency is one that **only exists in digital form**
- Currency is **stored in digital wallets** or accounts, there is **no physical bank notes** or **coins**
- It can be **exchanged online to pay for goods and services** or for transferring money internationally
- Unlike traditional currencies, **digital currency is not controlled or monitored** by central banks or government (**decentralised**)
- This can make digital currencies **highly volatile**, their value can fluctuate rapidly in a short period of time
- Digital currency **investment** can be risky due the volatility
- Digital currency can also be known as **crypto currency**
- All transactions made using crypto currency are **publicly available** and tracked using **cryptography**
- Examples include
 - **Bitcoin**
 - **Ethereum**



Examiner Tips and Tricks

In the IGCSE exam digital currency and crypto currency are the same thing, although in real life they are not strictly the same thing.

Blockchain

What is a blockchain?

- A blockchain is a **digital ledger** that records every transaction made with a particular digital currency
- Transactions are **time-stamped** and added to the blockchain in a **permanent** and **unalterable** way
- A blockchain is a **decentralised technology**, meaning that it is not controlled by a single entity or authority

- Instead, **every participant** in the network has **a copy of the ledger** and can **verify the transactions** independently
- The blockchain is made up of "**blocks**" of transactions that are **linked together** in a "chain" using **cryptographic algorithms**
 - This creates a **secure** and tamper-proof record of **every transaction** made with the digital currency
- Each transaction in the blockchain must be **verified by multiple participants** in the network
 - This verification process ensures that the transaction is **legitimate** and **prevents any fraudulent activity**



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